

Relationship of dietary protein and soy isoflavones to serum IGF-1 and IGF binding proteins in the Prostate Cancer Lifestyle Trial.



High levels of insulin-like growth factor 1 (IGF-1) are associated with increased risk of prostate cancer, whereas increased levels of some of its binding proteins (IGFBPs) seem to be protective. High intakes of dietary protein, especially animal and soy protein, appear to increase IGF-1. However, soy isoflavones have demonstrated anti-proliferative and apoptotic effects both in vitro and in vivo. We evaluated dietary intakes of total protein and soy isoflavones in relation to the IGF axis in prostate cancer patients making comprehensive lifestyle changes including a very low-fat vegan diet supplemented with soy protein (58 g/day). After one year, intervention group patients reported significantly higher intakes of dietary protein and soy isoflavones compared to usual-care controls ($P < 0.001$). IGF-1 increased significantly in both groups, whereas IGFBP-1 rose in the experimental group only ($P < 0.01$). Increases in vegetable protein over one year were associated with increases in IGFBP-1 among intervention group patients ($P < 0.05$). These results suggest that dietary protein and soy isoflavones, in the context of comprehensive lifestyle changes, may not significantly alter IGF-1. However, given the recent literature indicating that high intake of protein rich in essential amino acids (animal or soy protein) may increase IGF-1, it may be prudent for men with early stage prostate cancer not to exceed dietary protein recommendations.